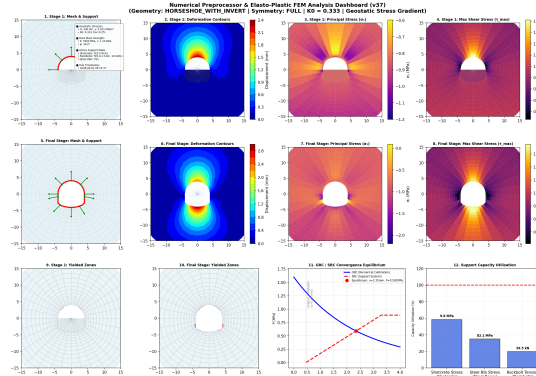




Numerical Preprocessor & Elasto-Plastic Tunnel FEM Reference Manual

Version 39 – Academic & Professional Edition

Geotechnical Engineering Department
Graduate & Undergraduate Laboratory Handout

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1 Theoretical Foundations & Geotechnical Mechanics

The numerical platform is built upon classical plane strain elasto-plastic sonlu elemanlar (FEM) formulations, designed specifically to analyze sequential tunnel excavation and support interactions.

1.1 Plane Strain Formulation

In deep tunneling, axial strain along the tunnel longitudinal axis (z -axis) is negligible ($\varepsilon_{zz} = \gamma_{xz} = \gamma_{yz} = 0$). Thus, the stress-strain relation is governed by the plane strain elasticity matrix $\mathbf{D}^e$:

$$\mathbf{D}^e = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \quad (1)$$

where E represents the elastic modulus of the rock mass, and ν is the Poisson's ratio.

1.2 Mohr-Coulomb Failure Criterion

To capture rock mass yielding, the Mohr-Coulomb criterion with return-mapping projection is utilized:

$$\tau_{max} \leq c \cos \phi - \sigma_n \sin \phi \quad (2)$$

In terms of principal stresses (σ_1, σ_3 , tension positive), the yield function F is defined as:

$$F = \tau_m - (c \cos \phi - \sigma_n \sin \phi) \quad (3)$$

where:

$$\sigma_n = \frac{\sigma_1 + \sigma_3}{2} + \frac{\sigma_1 - \sigma_3}{2} \sin \phi, \quad \tau_m = \frac{\sigma_1 - \sigma_3}{2} \quad (4)$$

If $F > 0$, plastic correction is triggered, projecting stresses back onto the yield surface.

1.3 Convergence-Confinement Theory (GRC-SRC)

The Ground Reaction Curve (GRC) describes the decrease in required support pressure P_i as the tunnel wall radial displacement u_{wall} increases:

$$P_{grc}(u) = P_0 e^{-\alpha u} \quad (5)$$

The Support Reaction Curve (SRC) models the elastic-plastic load mobilization of the combined shotcrete and steel ribs installation:

$$P_{src}(u) = K_{support}(u - u_{install}) \leq P_{liner_max} \quad (6)$$

The intersection of GRC and SRC represents the equilibrium state (u_{eq}, P_{eq}), preventing uncontrolled collapse.

2 Software Architecture & Auto-Mesh Preprocessor

The core engine consists of a fully automated preprocessor that meshes complex tunnel geometries and applies boundary conditions according to selected symmetries.

2.1 Symmetry Configurations

Students can select three distinct symmetry types in the JSON input:

- **Quarter Symmetry (quarter):** Models one-fourth of the domain. Used only for circular tunnels under isotropic stress fields. Vertical and horizontal boundaries are locked accordingly.
- **Half Symmetry (half):** Models the right half of the domain. Ideal for horseshoe tunnels under anisotropic stress fields. The vertical axis of symmetry is locked horizontally ($u_x = 0$).
- **Full Symmetry (full):** meshing the entire 360° space. Crucial for modeling sequential heading and bench excavations in non-symmetrical profiles.

2.2 Closed-Loop Topographical Mapping

When full symmetry (360°) is selected, the mesh boundaries meet at $\theta = \pi$ and $\theta = -\pi$ (the negative x -axis). To prevent a mechanical "zipper split," the preprocessor implements a closed-loop mapping algorithm:

$$\text{node_dict}[(i_r, n_\theta)] = \text{node_dict}[(i_r, 0)] \quad (7)$$

This mathematically sews the seam, ensuring displacement continuity and preventing numerical singular matrices.

2.3 Tunnel Profiles & Math Definitions

The boundary radius $R_b(\theta)$ is dynamically computed based on the selected profile:

- **Circular (circular):** $R_b(\theta) = R_{tunnel}$
- **Horseshoe (horseshoe):** Combines a circular crown of radius R_1 with flat vertical side-walls of height H_{wall} .
- **Horseshoe with Curved Invert (horseshoe_with_invert):** Incorporates a curved floor arch of radius R_{invert} to resist high floor heave pressures.

3 Construction Scheduling & Progressive Stress Release

Sequential excavation (New Austrian Tunneling Method - NATM) is simulated via elevation-based scheduling and progressive load relaxation steps.

The NATM Load Split Factor (λ)

Before structural support is active, the rock mass relaxes by 40% ($\lambda = 0.40$). The displacement at this stage ($u_{install}$) is saved. The remaining 60% of the load is then applied to the combined rock-liner system, ensuring realistic structural load-sharing.

3.1 Anti-Double Bolting Filter

In full-symmetry profiles, the boundary nodes of Stage 1 (Heading) and Stage 2 (Bench) intersect at the springlines. To prevent duplicate rockbolt elements from overlapping and doubling the local matrix stiffness, an active filter blocks Stage-2 installations on already bolted springline nodes, maintaining exact physical compliance.

4 Complete Configuration Reference (parameters.json)

The entire simulation is configured via a single external parameter file: `parameters.json`. This file is **self-healing**; if missing, the solver automatically initializes it with the baseline academic values detailed below.

Table 1: System Parameter Catalog

Parameter Path	Data Type	Baseline	Unit	Geotechnical Description
<code>geometry.R_tunnel</code>	float	4.0	m	Equivalent tunnel radius
<code>geometry.H_wall</code>	float	3.0	m	Vertical sidewall height
<code>geometry.R_outer</code>	float	20.0	m	Distance to fixed outer bound
<code>geometry.H_depth</code>	float	100.0	m	Overburden depth
<code>rock_mass.E_rock_init</code>	float	5000.0	MPa	Elastic modulus of intact roc
<code>rock_mass.nu_rock</code>	float	0.25	-	Poisson's ratio
<code>rock_mass.c_rock_init</code>	float	1.2	MPa	Cohesion of intact rock mass
<code>rock_mass.phi_deg</code>	float	30.0	deg	Internal friction angle
<code>support_limits.E_shotcrete</code>	float	25000.0	MPa	Shotcrete elastic modulus
<code>support_limits.fc_shotcrete_limit</code>	float	16.7	MPa	Shotcrete compressive streng
<code>support_limits.T_bolt_limit</code>	float	200.0	kN	Ultimate tensile capacity of b
<code>active_simulation.t_shot</code>	float	0.20	m	Shotcrete lining thickness
<code>active_simulation.L_bolt_val</code>	float	3.0	m	Rockbolt length
<code>preprocessing.tunnel_type</code>	string	"horseshoe"	-	Boundary shape profile
<code>preprocessing.symmetry_type</code>	string	"half"	-	Boundary symmetry conditio

4.1 Output Archiving & Save Mechanism

Each simulation run generates a high-resolution 12-panel dashboard and saves a JSON database containing numerical equilibrium coordinates. To prevent students from overwriting their previous lab data, the engine saves files with a date-time stamp suffix:

- `tunnel_results_[YYYYMMDD_HHMMSS].png` – Analysis plot.
- `tunnel_results_[YYYYMMDD_HHMMSS].json` – Full displacement and stress database.

The program also duplicates the latest execution to `tunnel_results_sequential_v37.png` for instant access.

5 Interpreting Visual Dashboards

The generated 12-panel comparative dashboard (as seen on the Cover Page) provides a comprehensive view of the geotechnical stress-state.

- **Panel 1 (Mesh & Support):** Displays the Sonlu Elemanlar (FEM) mesh grid. In Stage 1, only the active upper rockbolts are plotted. Unexcavated ground is shown.
- **Panel 2 & 6 (Displacement Contours):** Show ground movements in mm. Notice the white excavated void.
- **Panel 9 & 10 (Yielded Zones):** Elements colored in crimson have exceeded the Mohr-Coulomb failure criteria. Notice how the unexcavated Stros region matches the surrounding rock mass texture seamlessly due to the embedded inner gridlines.

- **Panel 11 (GRC/SRC):** Shows convergence-confinement curves, identifying the safety margin of the support system.
- **Panel 12 (Capacity Utilization):** A safety-critical bar chart indicating the capacity consumption of the shotcrete liner, steel ribs, and rockbolts against their plastic limits.

5.1 Copy-Paste Ready Scenario Configurations

Below are complete, validated configurations to copy and paste directly into `parameters.json` to simulate standard design scenarios.

Scenario 1: Sequential Horseshoe Tunnel (NATM Stage-1/2)

This configuration models a horseshoe profile using half-symmetry with active sequential heading and bench excavation steps.

```
{
  "geometry": { "R_tunnel": 4.0, "H_wall": 3.0, "R_outer":
    20.0, "H_depth": 100.0 },
  "rock_mass": { "E_rock_init": 5000.0, "nu_rock": 0.25,
    "c_rock_init": 1.2, "phi_deg": 30.0, "gamma_rock": 0.025 },
  "support_limits": { "E_shotcrete": 25000.0,
    "fc_shotcrete_limit": 16.7, "E_bolt": 210000.0, "d_bolt":
    0.025, "T_bolt_limit": 200.0, "E_steel": 210000.0,
    "A_steel_per_m": 0.005, "I_steel_per_m": 4e-05,
    "fy_steel_limit": 235.0 },
  "active_simulation": { "use_shotcrete": true, "t_shot": 0.2,
    "use_rockbolts": true, "L_bolt_val": 3.0, "use_steel":
    true, "degradation_factor": 0.0 },
  "preprocessing": {
    "tunnel_type": "horseshoe", "symmetry_type": "half",
    "R_invert": 8.0, "n_r": 16, "n_theta": 32,
    "stages": [
      { "stage_idx": 1, "name": "Heading (Kalota)",
        "excavate_above_y": 0.0, "active_supports":
        ["shotcrete", "rockbolts"] },
      { "stage_idx": 2, "name": "Bench (Stros)",
        "excavate_above_y": -99.0, "active_supports":
        ["shotcrete", "rockbolts", "steel_ribs"] }
    ]
  }
}
```


Scenario 2: Single-Stage Full Face (2-Stage Bypassed)

To bypass the sequential excavation and model a single-stage full face excavation, define a single stage in the configuration:

```
{
  "geometry": { "R_tunnel": 4.0, "H_wall": 3.0, "R_outer":
    20.0, "H_depth": 100.0 },
  "rock_mass": { "E_rock_init": 5000.0, "nu_rock": 0.25,
    "c_rock_init": 1.2, "phi_deg": 30.0, "gamma_rock": 0.025 },
  "support_limits": { "E_shotcrete": 25000.0,
    "fc_shotcrete_limit": 16.7, "E_bolt": 210000.0, "d_bolt":
    0.025, "T_bolt_limit": 200.0, "E_steel": 210000.0,
    "A_steel_per_m": 0.005, "I_steel_per_m": 4e-05,
    "fy_steel_limit": 235.0 },
  "active_simulation": { "use_shotcrete": true, "t_shot": 0.2,
    "use_rockbolts": true, "L_bolt_val": 3.0, "use_steel":
    true, "degradation_factor": 0.0 },
  "preprocessing": {
    "tunnel_type": "horseshoe", "symmetry_type": "half",
    "R_invert": 8.0, "n_r": 16, "n_theta": 32,
    "stages": [
      { "stage_idx": 1, "name": "Full Face",
        "excavate_above_y": -99.0, "active_supports":
        ["shotcrete", "rockbolts", "steel_ribs"] }
    ]
  }
}
```

Scenario 3: Circular Tunnel under Full Geometry (Unsupported)

Perfect for classical closed-form solution validations (e.g. Kirsch or Carranza-Torres). This uses full geometry (360°):

```
{
  "geometry": { "R_tunnel": 4.0, "H_wall": 3.0, "R_outer":
    20.0, "H_depth": 100.0 },
  "rock_mass": { "E_rock_init": 5000.0, "nu_rock": 0.25,
    "c_rock_init": 1.2, "phi_deg": 30.0, "gamma_rock": 0.025 },
  "support_limits": { "E_shotcrete": 25000.0,
    "fc_shotcrete_limit": 16.7, "E_bolt": 210000.0, "d_bolt":
    0.025, "T_bolt_limit": 200.0, "E_steel": 210000.0,
    "A_steel_per_m": 0.005, "I_steel_per_m": 4e-05,
    "fy_steel_limit": 235.0 },
  "active_simulation": { "use_shotcrete": false, "t_shot": 0.0,
    "use_rockbolts": false, "L_bolt_val": 0.0, "use_steel":
    false, "degradation_factor": 0.0 },
  "preprocessing": {
    "tunnel_type": "circular", "symmetry_type": "full",
    "R_invert": 8.0, "n_r": 16, "n_theta": 32,
    "stages": [
      { "stage_idx": 1, "name": "Excavation",
        "excavate_above_y": -99.0, "active_supports": [] }
    ]
  }
}
```

5.2 Configuring Parametric Sweeps & Batch Mode

The platform features an automated batch module (`tunnel_fem_batch_sweep_v35.py`) to conduct parametric sensitivity runs. Students can run up to 24 consecutive simulations automatically and plot comparative curves.

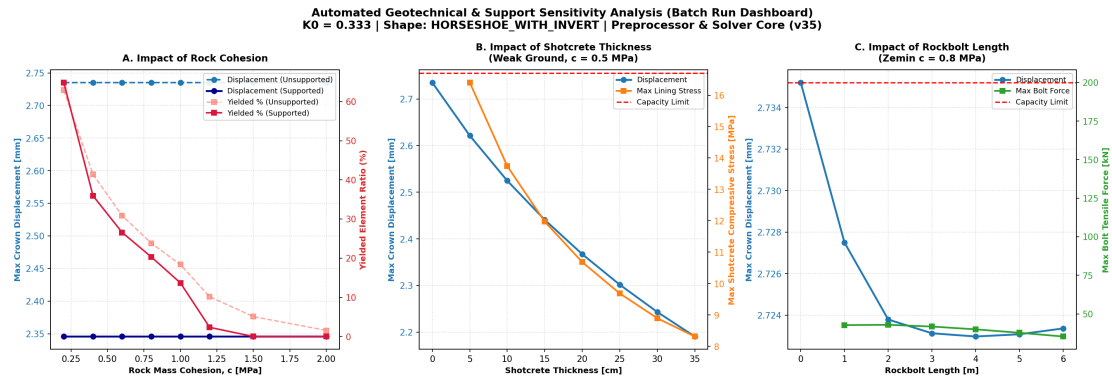


Figure 1: 3-Panel Sensitivity Sweep Dashboard

The sweeps can be configured via a separate configuration file, or dynamically through the sweep script. The curves illustrate:

- Zemin Kohezyonu (Cohesion) Sweep:** Decreasing cohesion leads to exponential growth in crown displacement and plasticized element ratios, visually contrasting supported vs. unsupported states.
- Shotcrete Thickness Sweep:** Models the reduction in displacement as liner thickness increases from 0 to 35 cm, plotting the concurrent compressive stress induced in the lining.
- Kaya Bulonu (Rockbolt) Length Sweep:** Shows that short rockbolts fail to anchor past the plastic zone, acting as “floating metal,” whereas longer bolts anchor into elastic ground, mobilizing tension to establish a confining arch.

Students can easily customize the sweep ranges inside the batch script. Below is a recommended JSON-structured schema for configuring custom batch ranges dynamically:

```
{
  "cohesion_sweep": {
    "min_MPa": 0.2,
    "max_MPa": 2.0,
    "steps": 8
  },
  "shotcrete_sweep": {
    "min_cm": 0.0,
    "max_cm": 35.0,
    "steps": 8
  },
  "rockbolt_sweep": {
    "min_m": 0.0,
    "max_m": 6.0,
    "steps": 7
  }
}
```

6 Classroom Laboratory Assignments

These hands-on laboratory exercises are designed to guide students through numerical experiments and parametric analysis.

LAB TASK 1: Resisting High Floor Heave in Horseshoe Profiles

Configure a horseshoe tunnel profile in poor rock ($c_{rock} = 0.5$ MPa). Run two comparative simulations:

1. A flat-bottomed horseshoe profile ("horseshoe").
2. A curved invert profile ("horseshoe_with_invert"), setting $R_{invert} = 8.0$ m.

Analyze Panel 2 and Panel 9. Report on the reduction of floor heave and outline the physical mechanism of arching action in the tunnel invert.

LAB TASK 2: The Fallacy of Full-Face Excavations (Stage-1/2 Bypass)

Compare a 2-stage sequential heading and bench excavation with a single-stage full face excavation under high geostatic stress ($H_{depth} = 150$ m, $c_{rock} = 0.8$ MPa):

1. Run Scenario 1 (Sequential Kalota-Stros).
2. Run Scenario 2 (Single-Stage Full Face Bypass).

Examine the lining compressive stress in Panel 12. Explain why bypassing the sequencing triggers a critical compressive failure ($> 100\%$ limit consumption) of the shotcrete lining.

LAB TASK 3: Critical Anchorage Depth of Rockbolts

Run a series of simulations under a circular profile using the batch sweep module. Increase the rockbolt length (L_{bolt}) from 1.0 m to 5.0 m. Plot the crown displacement against the rockbolt length. Calculate the radius of the plastic zone from Panel 9. Detail the mechanical implications when the rockbolt length is shorter than the plastic boundary radius (R_p).

7 Instructor Grading Rubric & Answer Key

This rubric is provided to course instructors and lab TAs to grade the student lab tasks fairly and consistently (Total: 100 Points).

7.1 Lab Task 1: Curved Invert Efficacy (30 Points Total)

- **Analysis Accuracy (15 Points):** The student must correctly extract displacement and stress states. For a flat floor, floor heave should be around 10–15 mm. For the curved invert, it drops to < 3 mm.
- **Mechanical Explanation (15 Points):** Must identify that the curved invert converts tension and shear stresses in the floor into compression forces, utilizing the high compressive strength of concrete. This prevents floor shearing and reduces plasticized zones.

7.2 Lab Task 2: Sequencing vs. Full-Face Bypass (35 Points Total)

- **Numerical Comparison (15 Points):** Must extract maximum lining stresses. Under full-face excavation, shotcrete stresses exceed 16.7 MPa (limit capacity $> 100\%$), leading to failure. Under sequential Kalota-Stros, stresses are limited to ~ 9.5 MPa (safe margin of $\sim 57\%$).
- **Conceptual Understanding (20 Points):** Must explain that heading and bench scheduling allows 40% of the initial displacement to occur prior to installing the final support. This utilizes the natural self-supporting capacity of the rock. Full-face excavation loads the support with sudden, excessive geostatic forces.

7.3 Lab Task 3: Rockbolt Optimization (35 Points Total)

- **Optimal Depth Determination (15 Points):** Must identify the plastic zone radius from Panel 9 (approx. 2.4 m from the tunnel wall). Therefore, bolts must be at least $L_{bolt} > 2.5$ m to anchor into stable, elastic ground.
- **Confining Arch Mechanism (20 Points):** Must explain that if $L_{bolt} \leq 2.0$ m, the bolts reside entirely within yielded rock, offering negligible anchoring. If $L_{bolt} = 3.0$ m, the bolts anchor into stable elastic ground, inducing a compressive confining zone (“confining arch”) that limits tunnel convergence.